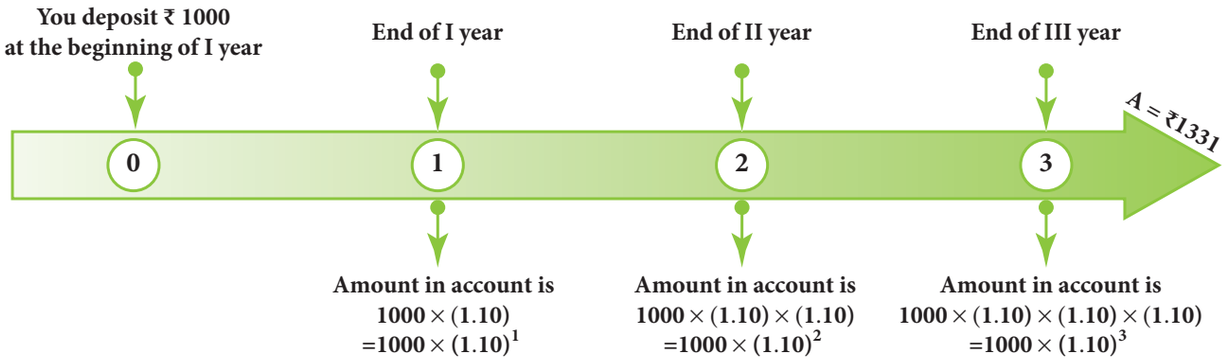
consuming. So, to save time and to find the amount and the compound interest easily we have a formula as explained in the following illustration.

Illustration 2

To calculate the amount and compound interest for ₹1000 for 3 years at 10% p.a compound annually.



Flow chart showing 10% interest being compounded annually

This leads to the pattern for Amount as $P\left(1+\frac{r}{100}\right)$ for the I year, $P\left(1+\frac{r}{100}\right)\left(1+\frac{r}{100}\right)$ for the II year, $P\left(1+\frac{r}{100}\right)\left(1+\frac{r}{100}\right)\left(1+\frac{r}{100}\right)$ for the III year and so on. In general, $A = P\left(1+\frac{r}{100}\right)^n$ for the nth year. Here, the amount after 3 years, $A = 1000\left(1+\frac{10}{100}\right)^3 = ₹1331$ and so, C.I = A – P = ₹331.



In 1626, Peter Minuit convinced the Wappinger Indians to sell him the Manhattan Island for 24 dollars. If the native Americans had put the 24 dollars into a bank account at 5% interest rate compounded monthly, by the year 2020 there would be well over 5.5 billion (550 crores) dollars in the account! This is the might of compound interest!



The following formulae will be helpful in calculating the compound interest easily for the following time periods.

(i) When the interest is compounded annually, we have

$$A = P\left(1+\frac{r}{100}\right)^n$$

where A is the amount, P is the principal, r is the rate of interest per annum and n is the number of years and we shall get the compound interest as C.I = Amount – Principal.

(ii) When the interest is compounded half yearly, we have

$$A = P\left(1+\frac{r}{200}\right)^{2n}$$

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